PHYSICS 2111 · PHYSICS NINJA PRACTICE EXAM

Exam 1 · Version B

20 questions · 90 minutes · original practice exam, not an official exam

Name _____ NetID _____ Section _____

Closed book. Calculators allowed. Use the official PHY 2111 formula sheet. Take $g = 9.81\ \text{m/s}^2$.
MC5 questions have five choices (6 points), MC3 three choices (3 points), MC2 two choices (2 points).
Mark one answer per question. Answers and worked solutions are in the separate solutions file.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A ball is thrown straight up at 15 m/s from a point 2 m above the ground. Ignore air resistance.

1) What is the maximum height of the ball above the ground? **MC5**

- **a.** 11.5 m
- **b.** 13.5 m
- **c.** 9.47 m
- **d.** 15 m
- **e.** 24.9 m

2) How long after it is thrown does the ball hit the ground? **MC5**

- **a.** 1.53 s
- **b.** 3.19 s
- **c.** 1.66 s
- **d.** 3.06 s
- **e.** 3.44 s

3) At the top of its path, the acceleration of the ball is **MC3**

- **a.** zero.
- **b.** 9.81 m/s² upward.
- **c.** 9.81 m/s² downward.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A car waits at a red light. The instant the light turns green, the car accelerates at 3 m/s², and at that same instant a truck passes it moving at a constant 20 m/s.

4) How long does it take the car to catch the truck? **MC5**

- **a.** 26.7 s
- **b.** 6.67 s
- **c.** 13.3 s
- **d.** 10 s
- **e.** 3.64 s

5) How fast is the car moving when it catches the truck? MC5

a. 60 m/s
b. 30 m/s
c. 26.7 m/s
d. 20 m/s
e. 40 m/s

6) At the instant the car is moving as fast as the truck, the car is MC2

a. behind the truck.
b. ahead of the truck.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A ball is launched from the roof of a 30 m tall building at 12 m/s, 30° above the horizontal. Ignore air resistance.

7) How far from the base of the building does the ball land? MC5

a. 47.3 m
b. 12.7 m
c. 32.8 m
d. 25.7 m
e. 37.9 m

8) How fast is the ball moving just before it lands? MC5

a. 12 m/s
b. 24.3 m/s
c. 31 m/s
d. 27.1 m/s
e. 36.3 m/s

9) If the ball were launched at 30° below the horizontal with the same speed, its speed just before landing would be MC3

a. the same.
b. larger.
c. smaller.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

An airplane flies at 250 km/h relative to the air. A wind blows toward the east at 60 km/h.

10) The pilot points the plane due north. What is its speed relative to the ground? MC5

- **a.** 310 km/h
- **b.** 190 km/h
- **c.** 257 km/h
- **d.** 243 km/h
- **e.** 250 km/h

11) To travel due north relative to the ground, the pilot should point the plane at what angle west of north? MC5

- **a.** 76.1 °
- **b.** 0 °
- **c.** 14.4 °
- **d.** 13.9 °
- **e.** 13.5 °

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 70 kg student stands on a bathroom scale in an elevator.

12) The elevator accelerates upward at 1.8 m/s². What does the scale read? MC5

- **a.** 813 N
- **b.** 126 N
- **c.** 705 N
- **d.** 561 N
- **e.** 687 N

13) Later the elevator is moving downward and slowing down. The scale reading is MC3

- **a.** equal to mg.
- **b.** less than mg.
- **c.** greater than mg.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A horizontal force of 36 N pushes a 5 kg block, which pushes a 7 kg block in front of it across a frictionless floor.

14) What is the magnitude of the force the 5 kg block exerts on the 7 kg block? MC5

- **a.** 0 N
- **b.** 21 N
- **c.** 36 N
- **d.** 28 N
- **e.** 15 N

15) If the same force pushed on the 7 kg block instead, the contact force between the blocks would be MC3

a. the same.
b. smaller.
c. larger.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 3 kg block slides down an incline at 35°. The coefficient of kinetic friction is 0.25.

16) What is the acceleration of the block? MC5

a. 5.63 m/s²
b. 3.62 m/s²
c. 2.45 m/s²
d. 7.64 m/s²
e. 5.58 m/s²

17) If the block were instead sliding up the incline, the magnitude of its acceleration would be MC3

a. smaller.
b. larger.
c. the same.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A car rounds a flat (unbanked) curve of radius 50 m. The coefficient of static friction between the tires and the road is 0.6.

18) What is the maximum speed at which the car can round the curve? MC5

a. 29.4 m/s
b. 17.2 m/s
c. 5.48 m/s
d. 22.1 m/s
e. 24.2 m/s

19) A heavier car on the same curve would have a maximum speed that is MC3

a. larger.
b. smaller.
c. the same.

THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.

A planet has half the radius of Earth and one eighth of Earth's mass. An astronaut weighs 600 N on Earth.

20) What does the astronaut weigh on the planet? **MC5**

a. 1200 N
b. 150 N
c. 75 N
d. 300 N
e. 600 N